

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA0202

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Comparative Analysis

Code of Conduct:

I CH.V.Ramayogi/192424319 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ch.V.Ramayogi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies	
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive	
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided	

Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated	
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand	

Annexure A

Comparative Analysis

1. CNNs vs. Vision Transformers (ViTs) for Object Detection Scenario: A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time using limited labeled datasets.

Questions:

1. Compare CNNs and Vision Transformers in terms of:

- **Feature extraction capabilities for local vs. global context**
- **Performance with small or sparse objects**
- **Computational complexity and inference speed for embedded systems**

2. Discuss which method is better suited for real-time deployment and how transfer learning or data augmentation could impact performance.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

Questions:

1. Compare PCA-based shape priors and graph-based learning in terms of:

- **Ability to handle non-rigid deformations**
- **Robustness to varying mesh resolutions and noise**
- **Ease of implementation and computational cost**

2. Analyze which approach would provide more accurate correspondences for realistic morphing and why.

ANSWERS:

1. CNNs vs. Vision Transformers (ViTs) for Object Detection

Factor	CNNs	Vision Transformers (ViTs)
Feature extraction	Excellent at extracting local features such as edges, corners, textures, and small object patterns. Deeper layers gradually capture global information.	Uses self-attention to capture global relationships between image regions from the beginning.
Small / sparse objects	Generally strong for small objects because convolution preserves local spatial information. Multi-scale CNN detectors can improve this further.	Can model relationships between distant regions, but small-object detection may require appropriate patch sizes, feature pyramids, and sufficient training data.
Limited labeled data	Usually performs well with smaller datasets, especially with transfer learning	Often more data-hungry when trained from scratch, although pretrained ViTs can perform very well.

	from pretrained CNNs.	
Computational complexity	Convolutions are highly optimized on GPUs and embedded hardware.	Self-attention can be computationally expensive, especially for high-resolution images, although efficient ViT variants reduce this cost.
Inference speed	Usually faster and more suitable for resource-constrained embedded systems.	May be slower and require more memory, depending on the architecture.
Real-time deployment	Very suitable for real-time applications such as autonomous vehicles.	Suitable when using lightweight/efficient transformer architectures and sufficient hardware.

Which is better for this scenario?

For a self-driving car that must detect **rare urban obstacles in real time with limited labeled data**, **CNN-based object detection is generally the safer choice.**

CNNs provide strong local feature extraction, which is important for detecting small objects such as traffic cones and animals. They also tend to have lower computational requirements and faster inference on embedded devices.

Transfer learning can significantly improve performance. A CNN pretrained on a large dataset can be fine-tuned using the company's smaller obstacle dataset. This reduces the amount of labeled data required.

Data augmentation can further improve robustness by applying:

- Rotation and flipping
- Scaling and cropping
- Brightness/contrast changes
- Blur and noise
- Weather and lighting variations

A lightweight CNN detector combined with transfer learning and targeted augmentation would therefore provide a good balance between **accuracy, speed, and limited training data**.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Factor	PCA-Based Shape Priors	Graph-Based Learning
Non-rigid deformation	Works well when shape variations can be represented by a limited number of principal components.	Better at learning complex and nonlinear deformations between mesh structures.
Varying mesh resolution	Usually assumes aligned or compatible representations, so different topology/resolution can be difficult to handle.	Can naturally exploit mesh connectivity and graph structure and can be designed to handle different resolutions.

Noise robustness	PCA can reduce some noise by retaining dominant components, but strong noise can affect the learned model.	Learned graph features can be made robust to noise, especially with suitable training data and regularization.
Implementation	Relatively simple: construct a shape matrix, calculate covariance, and perform PCA.	More complex because it requires graph construction, neural-network architecture, training, and correspondence learning.
Computational cost	Generally low after the PCA model is created.	Usually higher because graph-based neural networks require iterative computation.
Accuracy for complex shapes	Limited by the linear assumptions of PCA.	Better for complex, nonlinear shape variations.

Which approach is better for realistic morphing?

For the animation studio's problem, **graph-based learning would generally provide more accurate correspondences.**

Human faces undergo complex **non-rigid deformations** such as changes in facial expression, mouth movement, eye movement, and changes in facial geometry. PCA represents variation primarily through linear combinations of learned components, so it may struggle with large or highly nonlinear deformations.

Graph-based learning can use the **relationships between neighboring vertices and local geometric structures**. This makes it more appropriate for learning complex correspondences across facial meshes.

It can also be adapted to handle **different mesh resolutions and topology variations**, which is especially important in this scenario.

Final Conclusion

- **Self-driving obstacle detection:** CNNs are preferable when **real-time speed, limited labeled data, and embedded hardware** are major constraints.
- **3D face correspondence:** Graph-based learning is preferable when **high correspondence accuracy, non-rigid deformation, and varying mesh structures** are important.

Overall: CNNs prioritize efficient local feature extraction and fast inference, while graph-based learning provides greater flexibility for complex 3D geometric relationships.

